TEACH YOURSELF INTERMEDIATE XII UNIT 3

Programming Fundamentals (Advanced Python)

Collections, functions, and modules

What you will master

- Use lists, tuples, sets, and dictionaries appropriately
- Write functions with parameters and return values
- Handle files and exceptions at a basic level

How to use these notes

- Read one section, then close the book and explain it aloud.
- Do every *Try it* box before moving on.
- Memorize tables and truth patterns — exams love them.

3.1 Collections

```
marks = [88, 92, 75]          # list — mutable
coords = (30.2, 71.5)         # tuple — immutable
tags = {"python", "exam"}     # set — unique items
student = {"id": 1, "name": "Sara"}  # dict — key/value
```

Pick structure by need: ordered & changeable → list; fixed record → tuple; membership tests → set; lookup by key → dict.

3.2 Functions

```
def grade(m):
    if m >= 80: return "A"
    if m >= 60: return "B"
    return "C"

def average(nums):
    return sum(nums) / len(nums)
```

3.3 Files & errors

```
try:
    with open("data.txt") as f:
        print(f.read())
except FileNotFoundError:
    print("Missing file")
```

Teach Yourself — Practice & Check

1. Count word frequencies in a sentence using a dictionary.
2. Write a function is_palindrome(s) ignoring case.